

All Roads Lead to the Same Summit – But Who Draws the Map?

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1. From a physics homework

In a Physics 427 homework, I normalized pressure p , temperature T and volume V using the critical point, and rewrote the Van der Waals (VdW) equation without the material constants a and b :

$$\hat{p} = \frac{8\hat{T}}{3\hat{V} - 1} - \frac{3}{\hat{V}^2}.$$

If I know two of $\hat{p}, \hat{T}, \hat{V}$, I can determine the third without knowing a or b .

If I normalize using arbitrary values, a and b do not disappear. This is strange: “find the critical point” and “normalize to remove a and b ” seem to be on different levels, but they are tightly connected. I do not think this is a coincidence. There should be a framework. Under its description, the above 2 are equivalent.

2. Two descriptions

Algebraically: The first and second derivatives of $p(V)$ are zero if and only if

$$V_c = 3Nb, \quad k_B T_c = \frac{8a}{27b}, \quad p_c = \frac{a}{27b^2}.$$

Setting derivatives to zero gives the correct normalization. These two conditions must be special. They should correspond to the other description.

Physically: The critical point is where the phase transition ends. Different materials behave in a strikingly universal way here. If we use this universality as a measurement standard — a change of unit system — the results become independent of the material.

The phase transition ends at the point on the phase diagram where $p(V)$'s first and second derivatives are zero. So the two descriptions are unified harmonically.

If a pseudo-system has no critical point or loses universality near critical point, its $p(V)$'s derivatives cannot cancel a and b .

3. More General Algebra

I just made a correspondence between my algebraic operations and a physical explanation. But I believe I'm missing an algebraic framework that starts from the operations themselves, within which the disappearance of a and b follows directly.

By observing the mathematical form, we realize that we essentially want to express a and b using the normalized variables. We have the VdW equation, the two derivatives $= 0$. These three equations allow us to solve for $p_c(a, b)$, $V_c(a, b)$, and $T_c(a, b)$. This is a local map from the parameter space M to the variable space N : $(a, b) \mapsto (p_c, V_c, T_c)$. Once this map is a local bijection, a and b can be explicitly solved from p_c, V_c, T_c , and we succeed.

If we have more constants — $a, b, c, d, \dots$ — then T_c, p_c, V_c can still be expressed by those constants, but the constants may not be expressible from T_c, p_c, V_c . In that case, some constants will remain in the normalized equation.

Now comes the philosophical part.

4. Equifinality

We can explain the same phenomenon by physical or algebraic vision. Nothing restricts other visions. Consider the paths that explain a phenomenon as a net. As long as the logic is correct, you can reach the destination. This is self-consistency.

Do multi-path explanations mean that any phenomenon we have explained must have a new explanation that people have not found? This is not satisfying.

Do different visions correspond, or are they totally unrelated? Here, the physical explanation is that scaling at the critical point is independent of material constants. The algebraic explanation is that material constants can be expressed by normalized variables. These two do not correspond in their logical steps. This irrelevance also makes me uncomfortable.

These two discomforts push me to think further.

5. Dimensional collapse strike

I do not think any vision is more special than another. Thus I conjecture that there exists a higher-level vision that can include all of them — a meta-theory such that any theory is its translation or special case. This would also demonstrate self-consistency, because through a meta-theory every affiliated theory becomes related.

In the most general sense, logic itself might be the meta-theory. On a lower level, space-time structure is a good example: conservation, inertia and gravity are all determined by spacetime.

I give a conjecture consistent with my belief: *For a real, self-consistent phenomenon, there must be multiple paths to comprehend it. Different paths may differ greatly, but they will finally be unified under a higher framework.*

6. Paradigm, Method, Step

I need to distinguish three levels:

- **Paradigm:** different concepts or axiom systems (geometry / algebra / analysis).
- **Method:** within the same paradigm, different reasoning strategies (Newton / Taylor formula for binomial expansion).
- **Step:** within the same method, different orders of operations (expand then differentiate vs. differentiate then expand).

This article emphasizes Paradigm and Method.

7. Return to innocence

Fancy algebra always collapses back to the same result as physics. When I do algebra, I easily get lost in a sea of signs and figures. Sometimes I obtain extremely complicated expressions, but as long as I am guided by physics and make no mistakes, the complicated form always simplifies.

Suppose a person knows no physics and starts from the van der Waals equation doing aimless calculations. If lucky, he might find that a and b disappear. But a person who understands physics can simply start from physics and reach the same result.

Take climbing a mountain. There are different paths to the summit. One can climb rock by rock regardless of direction. He may wander, but theoretically he can still reach the

summit, because the summit exists and rocks are connected to it. But if he can see where the summit is, and find a creek to follow upstream, he will do much better.

In conclusion, algebra lacks intuition, but it is constrained by natural laws. Those constraints form a framework. Natural laws are so harmonious, and mathematical frameworks are so strikingly consistent with nature! Harmony comes from logic, and it is because of logic that nature can exist.

If nature exists first, and people find that nature fits the logic they artificially constructed after observing patterns. But why can nature not violate logic? Only by raising logic above nature can we resolve this. Moreover, any convincing human thought must be built on logic. Only when logic exists first can humans summarize patterns. Therefore logic must be the foundation of everything.¹

¹This essay originated from a reflection on a PHYS427 homework problem and has not been published elsewhere.